

Paper Practice 9.1**Exponential Functions**

WORKED KEY (odd-numbered items) – Show all of your work in the space provided. Round money to the nearest dollar unless told otherwise.

- 1.** Evaluate each function at the given value.

(Warm-Up Review)

a. $f(x) = 4^x$, when $x = 0$ ANSWER: 1

b. $f(x) = 3(2)^x$, when $x = 3$ ANSWER: $3 \cdot 8 = 24$

c. $f(x) = 5(\frac{1}{2})^x$, when $x = 2$ ANSWER: $5 \cdot \frac{1}{4} = 5/4 = 1.25$

- 2.** Determine whether each function shows exponential growth or decay. (Growth or Decay)

a. $y = 2(5)^x$

b. $y = 7(0.3)^x$

c. $y = 100(1.08)^x$

- 3.** Identify a and b for each exponential function $y = a(b)^x$. (Identify a and b)

a. $y = 6(3)^x$ ANSWER: $a = 6, b = 3$

b. $y = 0.5(\frac{1}{4})^x$ ANSWER: $a = 0.5, b = \frac{1}{4}$

- 4.** For $f(x) = 3(2)^x$, complete the table for $x = -1, 0, 1, 2, 3$. (Table)

- 5.** For $y = 5(2)^x$, state the domain, the range, the y -intercept, and the equation of the asymptote. (Key Features) ANSWER:
Domain: all real numbers. Range: $y > 0$. y -intercept: $(0, 5)$. Asymptote: $y = 0$.

- 6.** Determine whether the data shows exponential behaviour. Explain. (Exponential Behaviour)

a. x : 0, 1, 2, 3 y : 4, 12, 36, 108

- 7.** Determine whether the data shows exponential behaviour. Explain. (Exponential Behaviour)

a. x : 0, 1, 2, 3 y : 5, 9, 13, 17

ANSWER: No – the differences are a constant 4, so it is linear, not exponential.

- 8.** Each time you fold a piece of paper in half, the number of layers doubles. Start with 1 layer. (Apply)

a. Write a function for the number of layers after x folds.

b. How many layers after 7 folds?

- 9.** A car worth \$24,000 loses 15% of its value each year. (Apply)

a. Write an exponential function for its value after x years. ANSWER: $V(x) = 24000(0.85)^x$

b. Find its value after 3 years. ANSWER: $24000(0.85)^3 = 24000(0.614125) = \$14,739$

- 10.** A colony of 200 bacteria triples every hour. (Apply)

a. Write a function for the population after x hours.

b. How many bacteria after 4 hours?

- 11.** Error Analysis – Gus says $y = 3(0.5)^x$ is exponential growth because 3 is positive. Identify his mistake. (Error Analysis)

ANSWER: Growth or decay is decided by b , not a . Here $b = 0.5 < 1$, so it is decay.

- 12.** Explain what an asymptote is, and why the graph of $y = a(b)^x$ never touches the x -axis when $a \neq 0$. (Explain)